Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627. www

Date: _____ Class: _____ Instructor: _____ No. students _____ Observer Name: _____

Classroom (include any special notes about environment) _____

1. L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting, SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

2. Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3. L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

	1. Students doing														2. Instructor doing												3. Engagement				Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
0 - 2																															
2																															
4																															
6																															
8 - 10																															
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
10 - 12																															
12																															
14																															
16																															
18 - 20																															
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
20 - 22																															
22																															
24																															
26																															
28 - 30																															

Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www

Date: _____ **Class:** _____ **Instructor:** _____ **No. students** _____ **Observer Name:** _____

Classroom (include any special notes about environment) _____

1. L-Listening; **Ind**-Individual work; Fol - Follow-along worksheet; **CG**-Clicker Q discussn; **WG**-Worksheet group work; **OG**-Other group work; **AnQ**-Answer Q; **SQ**-Student Q; **WC**-Whole class discuss.; **Prd**-Predicting; **SP**-Student present.; **TQ**-Test/quiz; **W**-Waiting; **O**-Other
2. **Lec**-Lecturing; **RtW**-Writing; **FUp**-Follow-up; **PQ**-Pose Q; **CQ**-Clicker Q; **AnQ**-Answer Q; **MG**-Moving/Guiding; **1o1**-One-on-one; **D/V**-Demo+; **Adm**-Admin; **W**-Waiting; **O**-Other
3. **L** - low engagement (observer confident majority of students are not engaged); **M** - medium engagement (observer is unsure); **H** - high engagement (observer is confident majority of students are engaged); **Inter** - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
30 - 32																															
32																															
34																															
36																															
38 - 40																															
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
40 - 42																															
42																															
44																															
46																															
48 - 50																															

Ontime: 52 late |||| |||| || early-departure: 1

Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www

Date: 3/13 Class: 251 Instructor: Daniel No. students ____ Observer Name: Scott

Classroom (include any special notes about environment) ____

1. L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

2. Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3. L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

	1. Students doing														2. Instructor doing												3. Engagement				Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
0 - 2													X													X	X				started late (my fault)
2	X							X												X				X				X		1	
4	X														X									X				X			
6	X														X												X				
8 - 10	X														X												X				
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
10 - 12	X														X													X			
12	X				X										X	X		X											X		started VS activity
14					X																X	X							X	IIII	2 TAs moving around
16	X		X				X								X	X	X	X											X	I	
18 - 20	X		X												X	X													X		
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
20 - 22	X		X												X			X											X		was used to take notes on slide animation
22	X		X												X														X		
24	X		X												X			X											X	I	
26					X																X	X							X	IIII	
28 - 30	X				X												X				X								X		

Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education. Vol 12(4). pp. 618-627; ww

Date: _____ Class: _____ Instructor: _____ No. students _____ Observer Name: _____

Classroom (include any special notes about environment) _____

1. L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

2. Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3. L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
30 - 32	X														X														X		
32	X														X													X			
34	X		X												X	X													X		
36	X		X		X										X	X		X			X	X							X	11	
38 - 40					X											X	X				X	X				X			X		fring issue of polism
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
40 - 42	X		X				X									X	X	X											X		
42	X		X												X	X	X											X			
44	X							X							X					X								X			
46	X														X					X								X			Quiz ? on 5.10
48 - 50												X														X			X		survey

10am CDRLC 1476

3/4/ Date: _____ Class: 141 Instructor: Eda No. students ~50 Observer Name: Scott R.

Classroom (indude any special notes about environment) _____

L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	1. Students doing														2. Instructor doing													3. Engagement				Comments: explain difficult coding choices
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
0-2													X													X	X				↑
2													X													X	X				Instructor arrived at :03
4	X							X					X							X				X		X	X			1	began speaking at 10:05
6		X											X						X							X		X			Setting up slides/Clicker after a ?
8-10	X	X													X		X		X						X				X		pass-by-ref
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
10-12	X														X	X												X			Const
12	X														X	X		X									X				
14	X														X	X		X	X								X				
16	X	X													X	X	X	X	X										X		
18-20	X														X			X									X				
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd.	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
20-22	X														X	X												X			memory segments
22	X						X			X					X	X		X										X		1	code tracing prediction
24	X														X	X		X										X			
26	X														X	X											X				
28-30	X														X	X											X				

" Are there any ?'s ? " => all PQ

Date: _____ Class: _____ Instructor: _____ No. students _____ Observer Name: _____

Classroom (include any special notes about environment) _____

L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
30-32	X														X	X							X					X			
32	X						X								X			X					X					X		1	explain hw4 code issue (seg-fault)
34	X														X	X											X				heap allocs w/ new
36	X							X							X	X		X		X							X			1	
38-40	X														X	X											X				hep ex. dynamic
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
40-42	X							X							X	X		X		X								X		1	
42	X							X							X			X		X			X					X		1	Ads on online IDE distracting
44	X							X							X					X							X				
46	X							X							X			X		X							X			1	survey
48-50															X									X	X				X		

dismissed at 10:49:10

Att at start: 46 late: |||| |||| |||| | leave early _____

5 pm

Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www

Date: 3/11 Class: 211 Instructor: Hodges No. students _____ Observer Name: Scott

Classroom (include any special notes about environment) CORE 1426

1. L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

2. Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3. L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	1. Students doing													2. Instructor doing												3. Engagement				Comments: explain difficult coding choices
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
0 - 2	X												X	-										X	X			X			Starting ~30 sec. in
2	X														X												-	X			
4	X		X				X								X	X		X										X		1	
6	X		X												X	X													X		
8 - 10	X														X													X			
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
10 - 12					X																				X			X			Group vs. LL activity
12	X		X				X								X	X		X										X		1	
14	X		X				X								X	X		X										X		1	
16	X		X												X	X												X			
18 - 20	X		X				X		X						X	X		X										X		1	
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
20 - 22	X		X				X								X	X		X											X	1	
22	X		X				X	X							X	X		X		X								X		-	
24					X																				X			X			
26	X		X				X								X	X		X		X		X						X			
28 - 30	X		X												X	X		X									X				

Smith MK, Jones FHM, Gilbert SL, and Wieman CE 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS) a New Instrument to Characterize University STEM Classroom Practices CBE-Life Sciences Education, Vol 12(4), pp. 618-627, www

Date: _____ Class: _____ Instructor: _____ No. students _____ Observer Name: _____

Classroom (include any special notes about environment) _____

1, L-Listening; Ind-Individual work; FoI - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting, O-Other

2, Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3, L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	L	Ind	FoI	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
30 - 32	X														X													X			
32	X														X	X		X										X			"Any Q's"
34	X						X	X							X			X		X							X			1	
36	X						X								X			X				X						X		1	miscommunication during Q+A
38 - 40	X														X								X				X				
	L	Ind	FoI	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
40 - 42	X		X												X	X												X			
42	X		X					X							X	X												X			student asked to move sheet up on doc cam
44	X		X												X	X												X	X		
46	X													X	X								X			X	X				Survey
48 - 50	X													X										X		X					

Begin

Class end at 5:49

4pm

Students at start ~11.5 late |||| |||| leave early |||| |||

Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www

Date: 2/11 Class: 211 Instructor: Hodges No. students ____ Observer Name: Scott

Classroom (include any special notes about environment) CORLC 1426

1. L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

2. Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3. L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

	1. Students doing														2. Instructor doing											3. Engagement				Comments: explain difficult coding choices
min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
0 - 2	X												X											X				X			Started ~1 min in
2	X														X									X				X			exam return chat
4	X														X			X										X			
6	X		X				X								X	X		X									X			1	
8 - 10	X		X												X	X													X		
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
10 - 12	X		X										X		X	X												X			Slight audio adjustment
12	X		X												X	X												X			
14	X				X										X			X							X			X			
16	X				X		X								X	X		X										X		1	
18 - 20	X		X				X								X	X		X										X		1	
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
20 - 22	X		X												X	X												X			
22	X		X												X	X													X		
24	X		X												X	X											X				
26	X		X				X								X	X		X											X	1	
28 - 30	X		X												X	X		X										X			

37 47

40 60 40

Smith MK, Jones FHM, Gilbert SL, and Wieman CE. 2013. The Classroom Observation Protocol for Undergraduate STEM (COPUS): a New Instrument to Characterize University STEM Classroom Practices. CBE-Life Sciences Education, Vol 12(4), pp. 618-627; www

Date: _____ Class: _____ Instructor: _____ No. students _____ Observer Name: _____

Classroom (include any special notes about environment) _____

1. L-Listening; Ind-Individual work; Fol - Follow-along worksheet; CG-Clicker Q discussn; WG-Worksheet group work; OG-Other group work; AnQ-Answer Q; SQ-Student Q; WC-Whole class discuss.; Prd-Predicting; SP-Student present.; TQ-Test/quiz; W-Waiting; O-Other

2. Lec-Lecturing; RtW-Writing; FUp-Follow-up; PQ-Pose Q; CQ-Clicker Q; AnQ-Answer Q; MG-Moving/Guiding; 1o1-One-on-one; D/V-Demo+; Adm-Admin; W-Waiting; O-Other

3. L - low engagement (observer confident majority of students are not engaged); M - medium engagement (observer is unsure); H - high engagement (observer is confident majority of students are engaged); Inter - number of unique students instructor interacted with;

For each 2 minute interval, check columns to show what's happening in each category (or draw vertical line to indicate continuation of activity). Check multiple columns where appropriate.

COPUS

min	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
30 - 32					X													X										X			
32	X		X				X								X	X												X		1	
34	X		X												X	X		X											X		
36	X														X	X												X			
38 - 40	X														X	X		X									X				
	L	Ind	Fol	CG	WG	OG	AnQ	SQ	WC	Prd	SP	T/Q	W	O	Lec	RtW	Fup	PQ	CQ	AnQ	MG	1o1	D/V	Adm	W	O	L	M	H	Inter	
40 - 42	X						X								X			X									X			1	
42	X														X												X			1	
44	X		X												X	X												X			
46	X		X												X	X													X		
48 - 50	X		X											X	X	X								X					X		survey at end last 1 min